

GLOBAL CONFIDENCE REGIONS IN PARAMETRIC SHAPE ESTIMATION¹

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ABSTRACT

We introduce confidence region techniques for analyzing and visualizing the performance of two-dimensional parametric shape estimators. Assuming an asymptotically normal and efficient estimator for a finite parameterization of the object boundary, Cramér-Rao bounds are used to define a confidence region, centered around the true boundary. Computation of the probability that an entire boundary estimate lies within the confidence region is a challenging problem, because the estimate is a two-dimensional non-stationary random process. We derive lower bounds on this probability using level crossing statistics. The results make it possible to generate confidence regions for arbitrary prescribed probabilities. These global confidence regions conveniently display the uncertainty in various geometric parameters such as shape, size, orientation, and position of the estimated object, and facilitate geometric inferences. Numerical simulations suggest that the new bounds are quite tight.

1. INTRODUCTION

Several important imaging problems, including tomographic shape reconstruction, nonlinear inverse scattering, and computer vision, involve estimation of two-dimensional object shapes. We consider the following generic model for estimation of a shape s from N noisy data vectors $\mathbf{Y}_i$:

$$\mathbf{Y}_i \sim p(\mathbf{Y}_i|\mathbf{s}), \quad i = 1, \dots, N, \quad \mathbf{Y}_i \in \mathbb{C}^M, \quad (1)$$

where s is a two-dimensional boundary which describes the unknown *true* shape, and $p(\mathbf{Y}_i|\mathbf{s})$ is the observation model. This paper deals with parametric shape representations of the form

$$\mathbf{s}(t) \triangleq \mathbf{s}(t; \boldsymbol{\theta}) = \begin{bmatrix} s_x(t; \boldsymbol{\theta}) \\ s_y(t; \boldsymbol{\theta}) \end{bmatrix}, \quad t \in I, \quad (2)$$

where t is an index to points along the boundary, taking values in an interval $I = [0, T]$ for some fixed $0 < T < \infty$, $s_x(t; \boldsymbol{\theta})$ and $s_y(t; \boldsymbol{\theta})$ are the x and y coordinates of the boundary point indexed by t , and $\boldsymbol{\theta} \in \mathbb{R}^K$ is a K -dimensional parameter vector. The functions s_x and s_y are known functions, assumed to be bounded and continuously twice differentiable for all $\boldsymbol{\theta} \in \mathbb{R}^K$ and $t \in I$.

Parameterizations such as Fourier descriptors (FD) [1], B-splines, and wavelet descriptors [2] are special cases of this model and have been widely used for shape representation. In the shape estimation problem under the model

(2), the true vector parameter $\boldsymbol{\theta}$ is unknown, and the problem can be reduced to finding an estimate $\hat{\boldsymbol{\theta}}$ for $\boldsymbol{\theta}$ from the statistics

$$\mathbf{Y}_i \sim p(\mathbf{Y}_i|\boldsymbol{\theta}), \quad \boldsymbol{\theta} \in \mathbb{R}^K, \quad i = 1, \dots, N. \quad (3)$$

The boundary estimate is then $\hat{\mathbf{s}}(t) \triangleq \mathbf{s}(t; \hat{\boldsymbol{\theta}})$.

In practice, because $\mathbf{s}(t)$ describes the geometry of an object, one is more interested in assessing the quality of estimates of $\mathbf{s}(t; \boldsymbol{\theta})$ in easily interpreted geometric terms. Rather than the quality of estimates of $\boldsymbol{\theta}$ itself, what is needed is a global quality measure for the entire boundary $\{\hat{\mathbf{s}}(t), \forall t \in I\}$. For example, in a clinical radiotherapy application, an uncertainty estimate of the shape of a tumor would allow surgeons to effectively focus radiation on a tumor with prescribed probability, while minimizing unnecessary exposure to other healthy tissue. Likewise, in brain surgery, it is critical to avoid blood vessels. An uncertainty estimate of blood vessel boundaries may help neurosurgeons plan an optimum path for the insertion of a surgical instrument. Similar uncertainty quantification for a shape of an object would be useful in automated assembly lines, where robots solder and screw, and autonomous vehicles convey assembled parts by avoiding obstacles, all under guidance of an imperfect imaging system.

There are significant difficulties, however, in obtaining a geometrically meaningful global performance measure. While the estimate $\hat{\boldsymbol{\theta}}$ for unknown deterministic parameter $\boldsymbol{\theta}$ is a random vector, the estimate $\hat{\mathbf{s}}(t) = \mathbf{s}(t; \hat{\boldsymbol{\theta}})$ for unknown deterministic shape $\mathbf{s}(t; \boldsymbol{\theta})$ is a stochastic process continuously indexed by t . Performance bounds for $\hat{\mathbf{s}}(t)$ can be derived by extension of the Cramér-Rao inequality. Defining the shape estimation error for an unbiased estimator $\hat{\mathbf{s}}(t)$ by $\tilde{\mathbf{s}}(t) = \hat{\mathbf{s}}(t) - \mathbf{s}(t)$, $t \in I = [0, T]$, its mean-square-error $\mathbf{e}_s$ is bounded below by

$$\mathbf{e}_s = E \left[\frac{1}{T} \int_0^T \|\tilde{\mathbf{s}}(t)\|^2 dt \right] \geq \frac{1}{T} \int_0^T \mathbf{J}^{-1}(t, t) dt \quad (4)$$

where $\mathbf{J}(t, s)$ denotes the information kernel ([3] p.437-455). While the bound (4) for the shape estimate $\mathbf{s}(t; \hat{\boldsymbol{\theta}})$ is a useful criterion, it reduces information about an infinite dimensional random process into a single performance index $\mathbf{e}_s$. Pointwise bounds are also easy to derive:

$$E \|\tilde{\mathbf{s}}(t)\|^2 \geq \mathbf{J}^{-1}(t, t). \quad (5)$$

However, the bound (5) does not provide quantitative information about the entire boundary estimate $\{\mathbf{s}(t; \hat{\boldsymbol{\theta}}), \forall t \in I\}$

¹This work was supported by a grant from DARPA under Contract F49620-98-1-0498, administered by AFOSR.

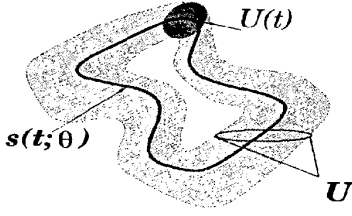


Figure 1: An example of local and global confidence regions, $\mathcal{U}(t)$ and $\mathcal{U}$, respectively, for a target boundary.

since it is derived for each individual point t , without considering the entire correlation structure of the random process $\hat{s}(t)$.

The main contribution of this paper is a technique for constructing small-size *global confidence regions* in which the entire boundary estimate lies with a given probability. These confidence regions can be conveniently visualized. (see Figure 1 for an example). They incorporate limits on the estimation performance for interesting geometric parameters such as shape, size, orientation, and the position of the object into an uncertainty band $\mathcal{U}$, which is created around the *true shape* $s(t; \theta)$. Then, one can investigate the fundamental performance of shape estimation from the geometric properties of the confidence regions.

In the formulation of confidence regions, a key parameter is the probability that the entire estimated boundary $\{s(t; \hat{\theta}), \forall t \in I\}$ lies in the confidence region. Its evaluation is a *level-crossing* problem [4]. In level-crossing terminology, our problem is interpreted as finding statistics that the stochastic process $s(t; \hat{\theta})$ stays between the barriers for all $t \in I$, where the barriers correspond to the inner and outer boundaries of the confidence region. The difficulty of the analysis is due to our confidence region problem having two-dimensional barriers, while all other conventional results for level-crossing by a scalar-indexed process are for one-dimensional barriers. Furthermore, even for periodic $s(t)$, the stochastic process $s(t; \hat{\theta})$ is non-stationary, so classical results for level-crossing of stationary random process cannot be applied.

In this paper, we derive lower bounds which appear to be quite tight. Such lower bounds are useful in practice, since one can generate somewhat larger confidence regions, and still guarantee the required global confidence level. In this paper, we also investigate the bounds for the classical definition of confidence regions for interval estimation [3]. Here, the confidence region is generated *a posteriori* around the estimate $s(t; \hat{\theta})$ (rather than around the true shape $s(t; \theta)$ as discussed above), and one is interested in finding the probability that the true shape $s(t; \theta)$ lies in the confidence region. We show that our lower bounds are also valid for this type of problem.

2. PROBLEM FORMULATION

Under appropriate regularity conditions [3], the maximum likelihood estimator (MLE) $\hat{\theta}$ of (3) is asymptotically unbiased, efficient and Gaussian-distributed (or best asymptotically normal (BAN)), and by the invariance property of the MLE, the asymptotic distribution of $s(t; \hat{\theta})$ is likewise

BAN: $s(t; \hat{\theta}) \sim \mathcal{N}(s(t; \theta), C_s(t))$ where the 2×2 covariance matrix $C_s(t)$ for $\hat{s}(t)$ is given by

$$C_s(t) = [\nabla_{\theta} s(t; \theta)]^T C_{\theta} [\nabla_{\theta} s(t; \theta)] \quad (6)$$

where $\nabla_{\theta} s(t; \theta) \in \mathbb{R}^{K \times 2}$ is the gradient of $s(t; \theta)$ with respect to θ , and the $K \times K$ covariance matrix $C_{\theta} = E[\theta - \theta][\theta - \theta]^T$ is the inverse of the Fisher information matrix. For a fixed t , according to classical estimation theory [5], a minimum size confidence region $\hat{\mathcal{U}}(t)$ for $s(t)$ at confidence level $\alpha \in [0, 1]$ is given by

$$\{x \in \mathbb{R}^2 : (x - \hat{s}(t))^T C_s(t)^{-1} (x - \hat{s}(t)) \leq \beta^2\} = \alpha, \quad (7)$$

for an appropriate $\beta > 0$ such that $\Pr\{\chi_2^2 \leq \beta^2\} = \alpha$, where χ_2^2 denotes the chi-square distribution with two degree of freedom. For each t , $\hat{\mathcal{U}}(t)$ is an ellipsoid centered on the boundary point $\hat{s}(t)$.

It is now desired to construct a global confidence region $\hat{\mathcal{U}}$ for the entire function $\{s(t), \forall t \in I\}$. A possible design is obtained by moving $\hat{\mathcal{U}}(t)$ along the boundary: $\hat{\mathcal{U}} \triangleq \bigcup_{t \in I} \hat{\mathcal{U}}(t)$. The region $\hat{\mathcal{U}}$ forms a “tube” around the estimated boundary $\hat{s}(t)$. Because $\hat{s}(t)$ is a random process, $\hat{\mathcal{U}}$ is a random set. The goal of this paper is to find a method to calculate the global confidence level for the so-formed confidence region $\hat{\mathcal{U}}$:

$$\gamma \triangleq \Pr\{s(t) \in \hat{\mathcal{U}}, \forall t \in I\}. \quad (8)$$

Another important probability of theoretical interest is

$$\Pr\{\hat{s}(t) \in \mathcal{U}, \forall t \in I\}. \quad (9)$$

where $\hat{s}(t)$ is the estimate of $s(t)$, and the deterministic confidence region $\mathcal{U}$ is defined as $\mathcal{U} \triangleq \bigcup_{t \in I} \mathcal{U}(t)$ where $\mathcal{U}(t)$ is given by

$$\{x \in \mathbb{R}^2 : (x - s(t))^T C_s(t)^{-1} (x - s(t)) \leq \beta^2\}, \quad (10)$$

Again, the region $\mathcal{U}$ is a tube, but this time centered around the true boundary. While the probability in (8) is the *a posteriori* probability that the true shape lies in a confidence region generated around the MLE $\hat{s}(t)$, the *a priori* probability in (9) focuses on predicting the fundamental uncertainty region for *any* asymptotically normal and efficient estimator. Therefore, in general they are different, and both are useful in practice.

The probabilities (8) and (9) are difficult to compute due to the overlaps of individual ellipses $\hat{\mathcal{U}}(t)$ (resp. $\mathcal{U}(t)$) for all $t \in I$. The goal of our paper is therefore to determine lower bounds on (8) and (9) that are reasonably tight. Such lower bounds are useful since one can generate somewhat larger confidence regions, and still guarantee the confidence levels.

3. MAIN RESULTS

The probabilities (8) and (9) satisfy the following bounds:

$$\Pr\{s(t) \in \hat{\mathcal{U}}, \forall t \in I\} \geq \Pr\{s(t) \in \hat{\mathcal{U}}(t), \forall t \in I\} \quad (11)$$

$$\Pr\{\hat{s}(t) \in \mathcal{U}, \forall t \in I\} \geq \Pr\{\hat{s}(t) \in \mathcal{U}(t), \forall t \in I\} \quad (12)$$

because there may exist some $t \neq t'$ such that $\mathbf{s}(t) \in \hat{\mathcal{U}}(t')$ while $\mathbf{s}(t) \notin \hat{\mathcal{U}}(t)$ (or $\hat{\mathbf{s}}(t) \in \mathcal{U}(t')$ while $\hat{\mathbf{s}}(t) \notin \mathcal{U}(t)$). Even though the probabilities in (8) and (9) are in general different, the right hand sides of (11) and (12) are identical. Indeed, the definitions (7) and (10) imply that for a given estimator $\hat{\mathbf{s}}$, the following events are the same:

$$\{ \mathbf{s}(t) \in \hat{\mathcal{U}}(t), \forall t \in I \} = \{ \hat{\mathbf{s}}(t) \in \mathcal{U}(t), \forall t \in I \}. \quad (13)$$

Due to the equivalence, in the sequel we derive the bounds for the left-hand side of (13) as a means to obtain lower bounds on the probabilities in (8) and (9).

3.1. Level-Crossing Bound

For convenience, we introduce the following notation:

$$\mathbf{Z} \triangleq \mathbf{C}_\theta^{-1/2} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \in \mathbb{R}^K \quad (14)$$

and

$$\mathbf{B}(t) = [\mathbf{B}_x(t) \quad \mathbf{B}_y(t)] = [\nabla_{\boldsymbol{\theta}} s_x(t) \quad \nabla_{\boldsymbol{\theta}} s_y(t)] \quad (15)$$

where the dependencies on $\boldsymbol{\theta}$ are implicit. It follows from the BAN of $\boldsymbol{\theta}$, that the transformed variable $\mathbf{Z}$ in (14) is normally distributed: $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Then, for all $t \in I$ and for all $\beta \in \mathbb{R}$, we have

$$\Pr \{ \mathbf{s}(t; \hat{\boldsymbol{\theta}}) \in \mathcal{U}(t) \} = \Pr \{ \mathbf{Z}^T \mathbf{P}(t) \mathbf{Z} \leq \beta^2 \}, \quad (16)$$

where $K \times K$ matrix appearing in (16) is:

$$\mathbf{P}(t) \triangleq \mathbf{C}_\theta^{1/2} \mathbf{B}(t) \left(\mathbf{B}(t)^T \mathbf{C}_\theta \mathbf{B}(t) \right)^{-1} \mathbf{B}(t)^T \mathbf{C}_\theta^{1/2}, \quad (17)$$

which is a rank-two projection on the range space of $\mathbf{C}_\theta^{1/2} \mathbf{B}(t)$. Applying the Gram-Schmidt orthonormalization procedure to the columns of $\mathbf{C}_\theta^{1/2} \mathbf{B}(t)$, $\mathbf{P}(t)$ is rewritten in terms of the resulting orthonormal vectors $\mathbf{v}_1(t)$ and $\mathbf{v}_2(t)$. The inequality of right-hand side of (16) can then be rewritten as $0 \leq R(t) \leq \beta$, where

$$R(t) \triangleq \sqrt{I_c(t)^2 + I_s(t)^2}, \quad t \in I \quad (18)$$

is the envelope process of the two scalar random processes

$$I_c(t) \triangleq \mathbf{Z}^T \mathbf{v}_1(t), \quad I_s(t) \triangleq \mathbf{Z}^T \mathbf{v}_2(t), \quad t \in I. \quad (19)$$

Then, (13) can be written as:

$$\Pr \{ 0 \leq R(t) \leq \beta, \forall t \in I \}. \quad (20)$$

Compared to the original Problem (9), Problem (20) is relatively simple, because for each $t \in I$, asymptotically $I_c(t)$ and $I_s(t)$ in (19) are independent, zero-mean, unit-variance Gaussian random processes, so $R(t)$ in (18) is a Rayleigh process, and the barrier of the level-crossing is reduced to a one-dimensional barrier.

However, the analysis is still complicated by the fact that $I_c(t)$ and $I_s(t)$ are non-stationary processes (hence $R(t)$ is non-stationary as well). Therefore, the classical results on level-crossing based on stationary assumptions [4] cannot be employed in our problem.

There is an analysis due to Shinozuka [6] for computing level-crossing probabilities, which we have found to be

useful for our problem. Following the notation in [6], define the probability

$$\begin{aligned} P[T; \beta] &\triangleq \Pr \left\{ \max_{t \in [0, T]} R(t) \geq \beta \right\} \\ &= 1 - \Pr \{ 0 \leq R(t) \leq \beta, \forall t \in [0, T] \}. \end{aligned} \quad (21)$$

Then, we have

$$P[T; \beta] \leq \int_0^T h(t, \beta) dt + P[0; \beta], \quad (22)$$

where $P[0; \beta]$ is the initial probability value at $t = 0$ and the function $h(t, \beta)$ is the *level-crossing rate* at t , defined by

$$h(t, \beta) = \int_0^\infty d\dot{R} p(\beta, \dot{R}; t) \quad (23)$$

where $\dot{R}$ denotes the derivative of R with respect to t and $p(\beta, \dot{R}; t)$ is the joint probability density function of R and $\dot{R}$ at $R = \beta$ and t [4]. The initial value $P[0; \beta]$ can be readily computed:

$$P[0; \beta] = \exp \left(-\frac{\beta^2}{2} \right), \quad (24)$$

since $R(0)$ is Rayleigh-distributed. Therefore, the problem of finding a lower bound for the probability in (13) is reduced to finding $h(t, \beta)$ for the non-stationary envelope process $R(t)$ of (18). Let

$$\mathbf{A} = \mathbf{C}_\theta^{1/2} \mathbf{B}(t) \mathbf{B}(t)^T \mathbf{C}_\theta^{1/2} \quad (25)$$

and denote by $\mathbf{A}^\dagger$ the pseudo-inverse of $\mathbf{A}$, and by $\mathbf{P}_\mathbf{A}^\perp$ the projection onto the null space of $\mathbf{A}$. In [7], we showed

Proposition 1

$$\Pr \{ \mathbf{s}(t; \hat{\boldsymbol{\theta}}) \in \mathcal{U}, \forall t \in I \} \geq 1 - (1 + \kappa \beta) \exp \left(-\frac{\beta^2}{2} \right),$$

where

$$\kappa \triangleq \int_0^T \left(\frac{\zeta(t)}{2\pi} \right)^{1/2} dt, \quad (26)$$

and $\zeta(t)$ is given by

$$\zeta(t) = \text{trace} \left\{ \mathbf{P}_\mathbf{A}^\perp \dot{\mathbf{A}} \mathbf{A}^\dagger \dot{\mathbf{A}} \mathbf{P}_\mathbf{A}^\perp \right\}. \quad (27)$$

4. NUMERICAL RESULTS

We now compute bounds for shape estimation in a practical scenario that arises in a passive radar imaging problem [8] and compare the results with Monte-Carlo simulations. In this problem, the target for shape estimation is parameterized by Fourier descriptors: $\mathbf{s}(t; \boldsymbol{\theta}) = \mathbf{B}(t)^T \boldsymbol{\theta}$, $t \in [0, T]$, where the matrix of basis function $\mathbf{B}(t)$ is given by

$$\mathbf{B}(t)^T = \begin{bmatrix} \mathbf{a}(t) & \mathbf{0} \\ \mathbf{r}(t) & \mathbf{c}(t) \end{bmatrix} \in \mathbb{R}^{2 \times K} \quad (28)$$

where $K = 4L + 1$ and

$$\begin{aligned} \mathbf{a}(t) &= [1 \cos(t) \cdots \cos(Lt) \sin(t) \cdots \sin(Lt)] \\ \mathbf{r}(t) &= \begin{bmatrix} \overbrace{0 \cdots 0}^{L+1} & \cos(t) & \overbrace{0 \cdots 0}^{L-1} \end{bmatrix} \\ \mathbf{c}(t) &= [1 \cos(2t) \cdots \cos(Lt) \sin(t) \cdots \sin(Lt)] . \end{aligned} \quad (29)$$

An example of shape $\mathbf{s}(t; \boldsymbol{\theta})$ parameterized by (28) is given in Figure 2 for $L = 7 (K = 29)$. The true parameter $\boldsymbol{\theta} \in \mathbb{R}^{29}$ and an example of $\mathbf{C}_{\boldsymbol{\theta}} \in \mathbb{R}^{29 \times 29}$ for Figure 2 can be obtained from [8]. An example of the deterministic confidence region $\mathcal{U}$ for Figure 2 with $\beta = 4$ is shown in Figure 3. In this example, thicker confidence regions are generated where the information conveyed by the scattered waves is insufficient, implying that shape estimation in that portion of the target is difficult.

The actual level-crossing probability

$$p_e = 1 - \Pr \left\{ \mathbf{s}(t; \hat{\boldsymbol{\theta}}) \in \mathcal{U}, \forall t \in I \right\} , \quad (30)$$

is computed by Monte-Carlo simulation, and is compared with the level-crossing bound. The results are shown in Figure 4. Monte-Carlo simulations were performed by drawing a random vector $\hat{\boldsymbol{\theta}}$ from a Gaussian distribution with mean $\boldsymbol{\theta}$ and covariance matrix $\mathbf{C}_{\boldsymbol{\theta}}$, generating the shape $\mathbf{s}(t; \hat{\boldsymbol{\theta}})$ and counting the events $\left\{ \mathbf{s}(t; \hat{\boldsymbol{\theta}}) \in \mathcal{U}, \forall t \in I \right\}$. The number of Monte-Carlo runs, N , is determined by $N \geq 1/(\eta^2 p_e)$ where η is a relative error level for the Monte-Carlo estimate (in this paper, we selected $\eta = 0.1$). The level-crossing bound for the error probability p_e is rewritten as $p_L = (1 + \kappa\beta) \exp(-\beta^2/2)$ so that $p_e \leq p_L$. Figure 4 also plots Monte-Carlo estimates of

$$p_{e2} = 1 - \Pr \left\{ \mathbf{s}(t; \hat{\boldsymbol{\theta}}) \in \mathcal{U}(t), \forall t \in I \right\} , \quad (31)$$

which, as discussed below (12), is an upper bound on p_e . Recall that our level-crossing bound is derived as an upper bound on (31). Figure 4 clearly confirms that our level-crossing bound is a tight bound on p_{e2} even for large K .

5. CONCLUSIONS

In this paper, we proposed a new concept for analyzing the performance of two-dimensional parametric shape estimators using a global confidence region. Cramér-Rao bounds for estimation of unknown geometric parameters such as shape, size, orientation, and the position of objects are used to construct a confidence region within which the shape estimate resides with a prescribed probability. The geometric properties of the global confidence region convey information about which parts of the shape are difficult to estimate and can be used to optimize imaging system parameters.

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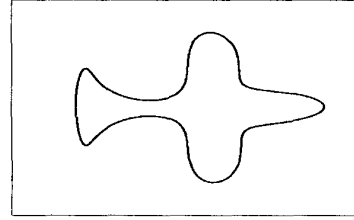


Figure 2: An example of shape $\mathbf{s}(t; \boldsymbol{\theta})$ parameterized by the Fourier descriptors of (28): $L = 7, K = 29$.

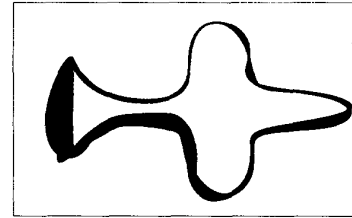


Figure 3: An example of confidence regions with $\beta = 4$ for the shapes of Figure 2: $L = 7, K = 29$.

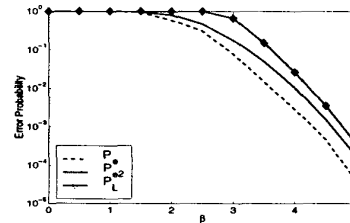


Figure 4: Comparison of Monte-Carlo simulation (p_e, p_{e2}) with level-crossing bound (p_L), for the shape given by Figure 2 ($K = 29$).